

ENTRIES 052-053 Capacity Push: 48 → 65 Physical Qubit Nodes — August 22, 2026 — Status: COMPLETE

## Entries 052-053 — Larger Star-GHZ Circuits, Wider Model Padding, and a Combined Re-Verification of All Prior Fixes

*Following Through on the Post-Review Roadmap: More Qubits, Same Rigor*

### 1. Motivation

Per project direction after Entry 051: rather than adding new features, grow the model's physical-qubit capacity from 48 nodes (Entry 045's padding) toward 60-70, to better reflect circuits a real research user might actually want to route. Star-topology GHZ circuits at  $k=10, 12, 14$  logical qubits (1 control +  $k-1$  targets) were generated on both Kyiv and Sherbrooke, reaching a new real maximum of 65 physical nodes and 99 edges after routing (vs. 42/56 previously) -- within the 60-70 target range.

### 2. A Genuine Simulator Scaling Wall, Documented Rather Than Hidden

$k=10$  star circuits produced 227 real two-qubit gates after routing, and Aer's `matrix_product_state` simulator measured roughly 0.25s per shot on circuits this deep -- making the usual 4096-shot run take on the order of 17 minutes, well past a single sandbox command. Shots were reduced per circuit size (`SHOTS_BY_SIZE = {10: 256, 12: 128, 14: 32}`) to fit the time budget. This is a real, larger precision tradeoff than earlier entries and is documented in `entry052_grow_bigger.py`'s docstring rather than applied silently -- these 12 large circuits carry more sampling noise than the rest of the dataset.

### 3. Rebuild and Retrain at the New Scale

`entry053_build_graphs.py` combined the original 2,288-circuit set, Entry 045's 29 bigger circuits, and Entry 052's 12 new ones into 2,329 total graphs. `MAX_N/MAX_E` were widened from 48/64 to 80/104 (headroom above the observed 65/99 max, following the same margin convention as Entry 045). The retrained model keeps every fix validated so far: per-chip normalization and no chip-identity leakage (Entry 048), and 5x floor-collapse loss up-weighting (Entry 051) -- so this run tests whether all three hold up together at the larger scale, not just the capacity increase in isolation.

### 4. Results: Capacity Increase Did Not Cost Accuracy

| metric             | same-chip (80/20 split) | v4.1 baseline |
|--------------------|-------------------------|---------------|
| MAE                | 1.36 pts                | 3.67 pts      |
| R <sup>2</sup>     | 0.975                   | —             |
| floor-collapse MAE | 1.57 pts                | 14.67 pts     |

| train → test             | MAE (Entry 051, 48-cap) | MAE (Entry 053, 65-cap) | fc MAE (051) | fc MAE (053) |
|--------------------------|-------------------------|-------------------------|--------------|--------------|
| Kyiv → Sherbrooke (cold) | 4.99 pts                | 4.59 pts                | 8.69 pts     | 9.67 pts     |
| Sherbrooke → Kyiv (cold) | 2.76 pts                | 2.96 pts                | 3.48 pts     | 4.15 pts     |

Same-chip accuracy at the new 65-node capacity (MAE 1.36, R<sup>2</sup> 0.975) matches Entry 047's original ablation baseline at the smaller 42-node capacity (MAE 1.23, R<sup>2</sup> 0.977) closely -- capacity growth did not degrade in-distribution fit. Cross-chip transfer held up as well: Kyiv-to-Sherbrooke MAE actually improved slightly (4.99 to 4.59, R<sup>2</sup> 0.708 to 0.769) and Sherbrooke-to-Kyiv stayed essentially flat (2.76 to 2.96, R<sup>2</sup> 0.884 to 0.891). Floor-collapse MAE regressed modestly in both directions (8.69 to 9.67, 3.48 to 4.15) -- plausibly because floor-collapse cases are a fixed small count (117) against a now-larger and more varied circuit population, diluting their relative weight even under 5x up-weighting. Still well below v4.1's own 14.67-pt floor-collapse error.

### Entries 052-053 — Conclusion

The three independently-developed fixes -- generalization (Entry 048), floor-collapse robustness (Entry 051), and now capacity (Entries 052-053) -- compose without materially undermining one another. The model now routes real circuits up to 65 physical qubits (vs. 48 previously) while keeping cross-chip R<sup>2</sup> in the 0.77-0.89 range and floor-collapse error

roughly 2-3x better than v4.1's own baseline in both directions.

5. Next Step

Per the user's roadmap: continue growing the general dataset toward 3,000 circuits (currently 932 in the separate bell-pair pool, not yet merged into this combined set), then fold that growth into a future capacity/generalization/floor-collapse re-check the same way this entry did. Real-hardware validation (Entry 049) remains paused per explicit prior direction and is not part of this phase of work.

Research Log — Index Addendum (Entries 052-053)

| Entry     | Title                                                           | Date         | Status   |
|-----------|-----------------------------------------------------------------|--------------|----------|
| Entry 052 | Larger Star-GHZ Circuits (k=10,12,14, 60-70 node target)        | Aug 22, 2026 | Complete |
| Entry 053 | Graph Rebuild + Combined Retrain at 65-Node/80-Padding Capacity | Aug 22, 2026 | Complete |

Scripts: entry052\_grow\_bigger.py, entry053\_build\_graphs.py, entry053\_train\_gnn.py, entry053\_cross\_chip.py. Data: quantumbridge\_data/entry052\_biggerghz\_dataset.json, entry053\_graph\_dataset.json, entry053\_gnn\_results.json, entry053\_cross\_chip\_results.json. Max circuit size grew from 42 to 65 physical nodes; model padding grew 48/64 to 80/104; all three prior fixes (generalization, floor-collapse weighting) confirmed compatible with the capacity increase.